

CHAIINGDA, Russian Federation

WMO: 247390

Lat: 62.283N

Lon: 119.850E

Elev: 308

StdP: 97.68

Time Zone: 9.00 (E09)

Period: 83-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-50.2	-47.7	-51.8	0.0	-50.0	-49.5	0.0	-47.6	4.2	-23.6	3.5	-24.9	0.1	180	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	29.0	18.1	26.4	17.0	24.0	15.5	19.5	27.1	18.0	24.6	16.5	22.2	1.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	12.3	22.2	15.3	11.3	20.7	14.1	10.4	19.6	57.2	27.9	52.1	24.9	47.6	22.4	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.3	4.4	3.5	DB	-52.6	31.7	2.7	1.7	-54.5	32.9	-56.1	33.9	-57.7	34.9	-59.6	36.2	(4)
(5)			WB	-52.6	21.0	2.7	2.1	-54.5	22.5	-56.1	23.7	-57.7	24.8	-59.6	26.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.0	-34.2	-28.3	-17.5	-6.6	5.2	13.1	16.1	12.7	5.0	-7.5	-23.6	-31.3	(6)
(7)		DBStd	19.02	8.55	8.26	8.11	6.70	5.33	4.28	4.42	3.73	4.45	6.64	8.70	9.57	(7)
(8)		HDD10.0	6985	1369	1072	853	497	167	16	4	15	161	543	1009	1279	(8)
(9)		HDD18.3	9639	1628	1306	1111	747	407	166	98	178	400	801	1259	1538	(9)
(10)		CDD10.0	428	0	0	0	0	18	108	193	99	10	0	0	0	(10)
(11)		CDD18.3	42	0	0	0	0	0	8	30	4	0	0	0	0	(11)
(12)		CDH23.3	667	0	0	0	0	7	168	424	67	1	0	0	0	(12)
(13)		CDH26.7	198	0	0	0	0	1	37	146	15	0	0	0	0	(13)
(14)	Wind	WSAvg	1.3	0.7	1.0	1.4	1.6	1.5	1.3	1.2	1.3	1.3	1.1	1.0	(14)	
(15)	Precipitation	PrecAvg	385	18	14	13	16	36	47	66	57	40	31	27	20	(15)
(16)		PrecMax	535	31	31	21	36	63	99	156	94	85	55	46	40	(16)
(17)		PrecMin	268	8	6	2	7	13	25	23	20	17	11	9	6	(17)
(18)		PrecStd	59	6	6	6	8	14	14	32	20	16	10	9	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-11.6	-6.8	5.1	12.0	24.3	29.9	32.1	29.0	22.8	8.7	0.5	-5.6	(19)
(20)			MCWB	-11.9	-8.3	1.5	4.7	13.2	17.2	19.4	19.8	13.8	3.9	-0.9	-6.1	(20)
(21)		2%	DB	-15.0	-9.9	1.4	8.6	21.1	27.3	30.5	24.8	18.3	5.5	-3.8	-8.7	(21)
(22)			MCWB	-15.3	-10.6	-1.6	2.6	10.9	15.8	19.3	16.5	11.4	2.9	-4.3	-9.1	(22)
(23)		5%	DB	-17.7	-13.0	-1.5	6.6	18.2	25.0	28.0	23.0	14.9	3.1	-6.5	-12.4	(23)
(24)			MCWB	-18.1	-13.5	-3.4	1.4	9.3	14.7	18.5	15.2	9.4	1.2	-6.9	-12.8	(24)
(25)		10%	DB	-21.1	-16.1	-4.5	4.3	15.1	22.7	25.2	20.8	12.2	1.3	-9.9	-16.5	(25)
(26)			MCWB	-21.3	-16.5	-6.0	0.1	7.9	13.9	17.0	14.4	7.8	-0.1	-10.3	-16.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-11.9	-8.1	2.0	5.0	14.3	18.3	21.5	19.6	14.3	4.7	-0.7	-7.0	(27)
(28)			MCDWB	-11.7	-6.8	4.5	10.7	23.7	27.5	28.7	28.2	21.7	7.3	0.2	-6.6	(28)
(29)		2%	WB	-15.3	-10.6	-1.3	3.1	11.6	17.0	20.3	17.6	12.1	3.3	-4.3	-9.3	(29)
(30)			MCDWB	-15.1	-9.8	1.1	7.9	19.5	24.9	28.0	23.0	17.6	5.6	-3.8	-9.0	(30)
(31)		5%	WB	-18.0	-13.5	-3.4	1.8	9.9	15.7	19.3	16.4	10.2	1.5	-6.9	-12.9	(31)
(32)			MCDWB	-17.7	-13.0	-1.5	5.9	16.6	23.1	26.5	21.2	14.1	2.8	-6.5	-12.6	(32)
(33)		10%	WB	-21.3	-16.5	-6.0	0.4	8.3	14.5	17.9	15.1	8.4	0.0	-10.3	-17.0	(33)
(34)			MCDWB	-21.1	-16.1	-4.6	4.0	14.9	21.3	24.2	19.6	11.6	1.2	-10.1	-16.7	(34)
(35)	Mean Daily Temperature Range		MDBR	8.9	11.9	14.5	13.7	11.7	13.4	13.1	13.1	10.2	8.2	10.1	8.8	(35)
(36)		5% DB	MCDBR	10.5	12.1	12.0	14.9	16.4	18.0	16.5	16.5	14.6	7.6	11.1	10.0	(36)
(37)			MCWBR	10.2	11.8	9.8	10.0	8.7	7.9	7.0	8.5	8.3	5.3	11.4	9.9	(37)
(38)		5% WB	MCDWB	10.0	12.1	11.5	13.5	14.4	15.4	14.9	13.8	13.9	7.3	11.4	9.9	(38)
(39)			MCWBR	9.8	11.6	9.6	9.1	8.4	7.7	7.0	7.7	8.2	5.5	11.4	9.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.220	0.259	0.290	0.331	0.371	0.404	0.488	0.439	0.327	0.277	0.215	0.190	(40)	
(41)		taud	1.942	2.062	2.095	2.125	2.276	2.292	2.017	2.125	2.455	2.337	1.980	1.873	(41)	
(42)		Ebn at Noon	432	688	807	847	844	818	722	726	782	682	431	267	(42)	
(43)		Edh at Noon	38	71	101	122	116	117	150	124	73	54	34	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.33	1.30	3.06	5.01	5.32	5.70	5.20	3.95	2.46	1.13	0.47	0.16	(44)	
(45)		RadStd	0.02	0.07	0.11	0.23	0.34	0.37	0.32	0.24	0.28	0.10	0.05	0.01	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47)	Regional (3 neighbors)	+0.72	N/A	N/A	N/A	N/A	-227	-265	N/A	N/A	(47)

Nomenclature: See separate page